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Echographic Evaluation of Choroidal Melanomas Treated with Ruthenium-106 Plaques: The Treatment Failures

Key Words

Choroidal melanoma
Ophthalmic echography
Ruthenium-106 plaque

Abstract

From July 1990 to October 1995 we treated 45 patients with choroidal melanomas using ruthenium-106 plaques. The population mean age was 59.2 ± 10.4 years. In this paper, the authors describe in detail their clinical management and follow-up timing. They also report on the echographic pattern change after treatment and the clinical history of some failure cases.

Introduction

Conservative treatment of choroidal melanomas with ruthenium-106 plaques has been introduced at the Clinica Oculistica dell'Università di Ferrara in July 1990.

In this study, we report the results of 5 years' experience that included 45 patients. Treatment failures observed will be analysed.

Echography plays an important role in the diagnosis of choroidal melanomas [1–3] as in the follow-up of patients receiving conservative treatment [4–6].

Patients and Methods

From July 1990 to October 1995, we treated 45 patients with choroidal melanomas. The population mean age was 59.2 ± 10.4 years; 62% were women.

Followed treatment indications, radiation dose, used plaques and surgical technique have previously been reported by our group in other communications [7, 8].

The mean thickness of treated tumours was 3.2 ± 1.2 mm. The patients' follow-up ranged from 1 to 63 months.

Results

Tumour regression was observed echographically in the majority of the patients (87%). We describe it as follows: increased tumour reflectivity and reduction of maximal tumour thickness in standardized A scan; increased echogenicity, diameter reduction and reduction or disappearance of choroidal excavation in B scan as shown in figure 1a–d.

Patient condition is no cause for worry when ocular echographic ophthalmoscopic and fluorescein-angiographic signs, together with normal patient examination, blood tests and liver echography, are normal. In those cases, patient follow-up is continued at regular intervals.

Out of the 45 patients treated, 2 were lost to follow-up, 1 died for causes not related to the primary neoplasm and 6 were treatment failures.

One unclear and 5 treatment failure cases shall now be analysed.

Case 1

P.E., female, 68 years old, observed in January 1992 with a little melanoma across the superior temporal retinal vessels (OS). Considering tumour position and di-

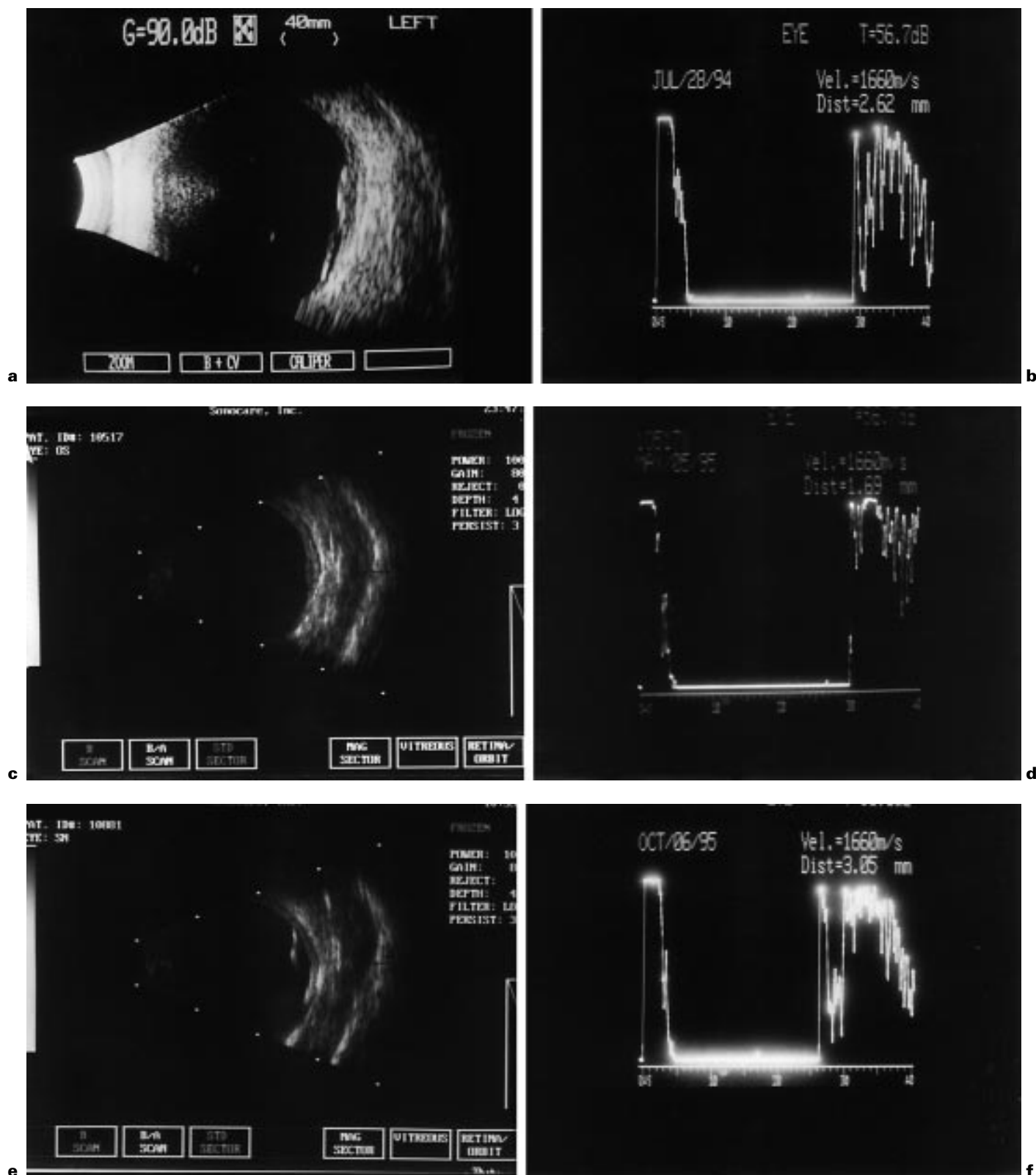


Fig. 1. **a, b** Choroidal melanoma before the ruthenium-106 treatment. **c, d** The same tumour 9 months after the ruthenium-106 treatment. **e, f** The same tumour 14 months after the ruthenium-106 treatment.

mension (2 mm in thickness), a conservative treatment with laser photocoagulation was chosen. After some time, this lesion showed an increase in thickness and focal signs of activity. So, we decided to use a ruthenium-106 plaque in July 1992. However, this treatment did not show the expected results either, while the mass increased in thickness. At this point, after having correctly informed the patient, we performed an enucleation in May 1994. The histological examination confirmed the presence of active mixed choroidal melanoma. So, neither the laser nor the plaque treatment modified the tumour.

Case 2

F.T., female, 64 years old, observed in September 1993 presenting melanoma in OD with a maximum of 3.35 mm in thickness. A treatment with a ruthenium-106 plaque was decided and then performed in September 1993. The patient was regularly followed; at all controls, the lesion did not show any ophthalmoscopic, fluorescein-angiographic or echographic modifications; on the contrary, after 9 months from treatment the lesion appeared grown in thickness. We decided to wait up to 12 months; as we saw a constant increase in thickness (4.32 mm), we decided to repeat the treatment in September 1994. The patient was regularly kept in follow-up until September 1995, but not having any response after the second treatment, we advised the enucleation following neoplastic growth. The patient refused surgery and is actually being followed.

Case 3

M.A., male, 65 years old, observed in July 1994 presenting a little melanoma of 2.62 mm in thickness (OS). The treatment was made in August; the patient was regularly followed up to May 1995, at 8 months from surgery, when a marked reduction in thickness and an enhancement in internal reflectivity were noted. This seemed a very good and rapid response to therapy. During the following control, in September 1995, we observed a radically modified picture: an increase in thickness (3.05 mm) and a clear reduction in internal reflectivity of the lesion (fig. 1a–f). In October 1995, after progression of the tumour towards the optic nerve had been evidenced, we advised enucleation. In November 1995, the patient was operated. In this case too, the histological examination revealed some zones of activity of the tumour; the type was mixed.

Case 4

T.F., male, 46 years old, observed in November 1994 with a melanoma in OS (5.51 mm in thickness). In the

same month we performed surgery. The patient was followed up for 1 year, but we did not observe any modification of the lesion, except for a moderate increase in thickness. For this reason, we advised the patient a new treatment with charged particle radiotherapy in another centre or an enucleation. The patient chose the first solution and so he was sent to another hospital.

Case 5

L.F., male, 65 years old, observed in March 1991 with a melanoma in OD (5.81 mm in thickness). In the same month, we performed the treatment; then, he was regularly followed for 44 months. In this period, we observed a decrease in thickness (3.32) and an increase in internal reflectivity. Even if these modifications appeared slowly, they were positive therapeutic signs. In February 1995 (almost 5 years from the treatment), the echographic picture seemed modified, since a modest increase in thickness (3.95 mm) and a decrease in internal reflectivity were observed. So, we decided for regular controls. In June 1995, another increase in thickness (4.95 mm) and decrease in internal reflectivity, signs of reactivation of the tumour, were observed. For this reason, we advised an enucleation which was performed in July 1995. The histological examination revealed zones of scar and activity together, the type was mixed.

Case 6

S.D., male, 29 years old, observed in October 1993 with a melanoma in OD (5.6 mm in thickness). Treated in the same month, he was kept in follow-up for 1 year with an apparent positive response to treatment. In January 1995, the echographic pictures seemed satisfactory, but signs of activity were present in fluorescein angiography. In this case too, we decided for regular controls. In October 1995, fluorescein-angiographic and orbital echographic pictures were evaluated. In them, a scleral anechogenic zone next to the lesion in the B scan and a scleral scaled signal near the tumour in the A scan were observed. At this point our diagnostic doubt was that of scleromalacia, related to the radiations, or an extrabulbar extension of the tumour. At present, the patient is undergoing further investigations.

Discussion and Conclusions

In more than 5 years of experience in the treatment of choroidal melanomas with ruthenium-106 plaques, the analysis of our first failures shows 13% of cases in which

radiotherapy did not yield the expected effects. However, this result remains in any case inferior to those of larger series in which percentages of failure or metastasis are surely higher [9–12]. Unfortunately, these series, not having a high number of patients with follow-up superior to 5 years, are not quite correlatable.

In all cases analysed, for the first year following the treatment our point of view was always that of waiting, as referred by Lommatzsch [10]: it is always necessary to expect the results after a long time; it is possible to talk of failure only if there have been no modifications of the tumour or there has been a growth after 1 year of observation.

Moreover, with this study we point out how radiotherapy does not always make any clinically clear modifications. This occurs also after 2 consecutive treatments, like

in 1 of our cases. In other cases, on the other hand, there is an initial positive response followed by growth after an interval which is variable (also 4 years).

Considering these failures, we must remember that both the kinetic of growth and the regression of irradiated choroidal melanoma are individually extremely variable; so, in the case of absent response to therapy, only an accurate follow-up will allow us to operate in time.

Furthermore, without any doubt we confirm that the primary method for a correct and secure follow-up is the echography, which is the most specific, sensible and non-invasive examination. Surely, ophthalmoscopy and fluorescein angiography too are very useful and, probably in the near future, colour Doppler sonography with echographic contrast medium could be positively used together with these methods.

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